Assigning ancestry proportions

2025-05-16

Load gt

```
mad_bor_g1000_sum <-
  gt_load("../data/intermediate/rbind_data/mad_bor_g1000_sum_clean.gt",
    reattach_to = "../data/intermediate/rbind_data/mad_bor_g1000_sum_clean"
  )
mad_bor_g1000_sum <- mad_bor_g1000_sum %>% group_by(population)
```

Load fastmixture results

```
qmats <-
  readRDS("../data/intermediate/fastmixture/fastmixture_m.RData")
# check order
testthat::expect_equal(qmats$id, mad_bor_g1000_sum$id)
# add grouping variable
qmats$group <- mad_bor_g1000_sum$population
```

Add continental region

```
mad_bor_g1000_sum$Region <- "Other"
mad_bor_g1000_sum[mad_bor_g1000_sum$population %in% c("Borneo", "Sumba", "Timor"), "Region"] <- "Island
mad_bor_g1000_sum[mad_bor_g1000_sum$population %in% c("BEB", "PJL"), "Region"] <- "South Asia"
mad_bor_g1000_sum[mad_bor_g1000_sum$population %in% c("CHS", "KHV"), "Region"] <- "East Asia"
mad_bor_g1000_sum[mad_bor_g1000_sum$population %in% c("ESN", "GWD", "LWK", "MSL", "YRI"), "Region"] <-
mad_bor_g1000_sum[mad_bor_g1000_sum$population %in% c("Betsimisaraka", "Merina", "Diego", "Sakalava Tsi
```

```
qmats$continents <- mad_bor_g1000_sum$Region
```

Find .Q values that represent African ancestry

The function calculates the average of each column in the Q matrix for all individuals within each population. Then, it subsets down to just the African populations. If the average is above 50% in any African population, it is determined to be an 'African' column. For cases where only one column is 'African' (k values 2-5 say) it assigns Merina individuals to African or Non-African depending on whether they are above or below 50%. For cases where more than one column is 'African' (K values > 5) it does a rowSum across all columns representing African ancestry, and then if the total is $> 50\%$ it assigns individuals as African.

```
merina_df <- data.frame(mad_bor_g1000_sum[,c(1,2)])
# create a dataframe of only merina individuals with "id" and "population" columns
merina_df <- merina_df %>% filter(population == "Merina")
```

```r
# The function takes qmats - the gt_admix object containing a list of all Q matrices
# and a dataframe of merina individuals with "id" and "population" columns

assign_african_ancestry <- function(qmats, merina_df){

  # Initialise a list to store which columns represent african ancestry for each run of each K
  assigned_african <- list()

  for(this_k in 2:max(range(unique(qmats$k)))){ # for each k
    # list within a list - so we have $K containing runs 1 2 3
    assigned_african[[as.character(this_k)]] <- list()
    for(this_rep in 1:rle(qmats$k)$lengths[1]){ # for each run
      # get the q_matrix
      k_run <- get_q_matrix(qmats, k = this_k, run = this_rep) %>% as.data.frame()
      k_run$id <- qmats$id # add id
      k_run$group <- qmats$group # add population
      k_run$continents <- qmats$continents # add the continent

    assigned_african[[as.character(this_k)]][[this_rep]] <- k_run %>%
      # group to calculate within each population
      group_by(group) %>%
      # get the mean of each column
      reframe(across(starts_with(".Q"), mean), continents = unique(continents)) %>%
      # filter to African populations
      filter(continents == "Africa") %>%
      # find whether mean .Q is over 0.5 in any African population
      summarise(across(starts_with(".Q"), ~ any(.x > 0.5))) %>%
      # pivot to long format
      pivot_longer(everything(), names_to = "Q_value", values_to = "flag") %>%
      # filter to those that are TRUE
      filter(flag) %>%
      pull(Q_value) # extract their column names into the assigned_african list

    # get the column names of the .Q values that represent African ancestry
    Q <- assigned_african[[as.character(this_k)]][[this_rep]]

    # if there is only one column
    if(length(Q)==1){
      # create column name from the current K and run
      column <- paste0(paste0("k",this_k),paste0("_r",this_rep))
          merina <- k_run %>%
            filter(group == "Merina") %>% # filter to merina
            # create a new column labelling individuals as African or Non-African
            mutate(
              !!column := ifelse(.data[[Q]] > 0.5, "African", "Non-African")
            ) %>%
            select(column)
          merina_df <- cbind(merina_df, merina)

    # if there are two or more columns
    } else if(length(Q)>1){
      # create column name from the current K and run
      column <- paste0(paste0("k",this_k),paste0("_r",this_rep))
```

```
        merina <- k_run %>%
          filter(group == "Merina") %>% # filter to merina
          # sum the across all columns representing African ancestry for each individual
          mutate(sum_q = rowSums(across(Q))) %>%
          # create a new column labelling individuals as African or Non-African
          mutate(
              !!column := ifelse(sum_q > 0.5, "African", "Non-African")
            ) %>%
            select(column)
        merina_df <- cbind(merina_df, merina)
      }
    }
  }
  return(as.data.frame(merina_df))
}

res <- assign_african_ancestry(qmats, merina_df)
```

```
## Warning: Using an external vector in selections was deprecated in tidyselect 1.1.0.
## i Please use `all_of()` or `any_of()` instead.
##   # Was:
##   data %>% select(column)
##
##   # Now:
##   data %>% select(all_of(column))
##
## See <https://tidyselect.r-lib.org/reference/faq-external-vector.html>.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```

```
## Warning: There was 1 warning in `mutate()`.
## i In argument: `sum_q = rowSums(across(Q))`.
## Caused by warning:
## ! Using an external vector in selections was deprecated in tidyselect 1.1.0.
## i Please use `all_of()` or `any_of()` instead.
##   # Was:
##   data %>% select(Q)
##
##   # Now:
##   data %>% select(all_of(Q))
##
## See <https://tidyselect.r-lib.org/reference/faq-external-vector.html>.
```

Results show inconsistency, a hardcoded 50% threshold for a .Q proportion to be determined as 'African' is not ideal.

```
res[20:26,]
```

```
##        id population       k2_r1       k2_r2       k2_r3       k3_r1
## 20 CH20RR     Merina Non-African Non-African Non-African Non-African
## 21 CH21RR     Merina     African     African     African     African
## 22 CH22RR     Merina     African     African     African     African
## 23 CH23RR     Merina     African     African     African     African
## 24 CH24RR     Merina     African     African     African     African
```

```
## 25 CH25RR     Merina     African     African     African     African
## 26 CH26RR     Merina     African     African     African     African
##          k3_r2       k3_r3       k4_r1       k4_r2       k4_r3       k5_r1
## 20 Non-African Non-African Non-African Non-African Non-African Non-African
## 21     African     African     African     African     African Non-African
## 22     African     African     African     African     African Non-African
## 23     African     African     African     African     African Non-African
## 24     African     African     African     African     African Non-African
## 25     African     African     African     African     African Non-African
## 26     African     African     African     African     African Non-African
##          k5_r2       k5_r3       k6_r1       k6_r2       k6_r3       k7_r1
## 20 Non-African Non-African Non-African Non-African Non-African Non-African
## 21 Non-African Non-African     African     African     African     African
## 22 Non-African Non-African     African     African     African     African
## 23 Non-African Non-African     African     African     African     African
## 24 Non-African Non-African     African     African     African     African
## 25 Non-African Non-African     African     African     African     African
## 26 Non-African Non-African     African     African     African     African
##          k7_r2       k7_r3       k8_r1       k8_r2       k8_r3       k9_r1
## 20 Non-African Non-African Non-African Non-African Non-African Non-African
## 21 Non-African     African Non-African Non-African Non-African Non-African
## 22 Non-African     African     African     African     African     African
## 23 Non-African     African Non-African Non-African Non-African Non-African
## 24 Non-African     African Non-African Non-African Non-African Non-African
## 25 Non-African     African Non-African Non-African Non-African Non-African
## 26 Non-African     African Non-African Non-African Non-African Non-African
##          k9_r2       k9_r3      k10_r1      k10_r2      k10_r3
## 20 Non-African Non-African Non-African Non-African Non-African
## 21 Non-African Non-African Non-African Non-African Non-African
## 22     African     African     African     African     African
## 23 Non-African Non-African Non-African Non-African Non-African
## 24 Non-African Non-African Non-African Non-African Non-African
## 25 Non-African Non-African Non-African Non-African Non-African
## 26 Non-African Non-African Non-African Non-African Non-African
```